



MULTIMEDIA UNIVERSITY OF KENYA

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

UNIVERSITY EXAMINATIONS 2023/2024

SECOND YEAR SUPPLEMENTARY EXAMINATION FOR THE DEGREE OF

BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND BACHELOR OF

SCIENCE IN SOFTWARE ENGINEERING

CCS 2216: INTEGRAL CALCULUS

DATE: 31ST JULY 2024

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

- a) Define an indefinite integral.
- b) Evaluate the following using substitution method
 - i) $\int (x^3 + x)^5 (3x^2 + 1) dx$ (3marks)
 - ii) $\int \cos(7\theta + 3) d\theta$ (3marks)
- c) *T-method* is one of the techniques used in integration.
 - i) Express and simplify in terms of t : $\sec x + \tan x$ (2marks)



- ii) Evaluate using *t*-method: $\int \frac{1}{1+\sin x + \cos x} dx$ (3marks)
- iii) State the formula for volume by disk method for a rotation about the *y*-axis (1mark)
- iv) Find the volume of the solid generated by revolving the region between the parabola $x = y^2 + 1$ and the line $x = 3$ about the line $x = 3$. (3marks)
- d) (i) Define an arc length, *L*, for the curve $y = f(x)$ from a point $A = (a, f(a))$ to a point $B = (b, f(b))$ (1mark)
- (ii) Find the length of the arc of the curve from point *P* to point *Q* if $y = \frac{1}{2}x^2$, $P(-1, \frac{1}{2})$, $Q(1, \frac{1}{2})$ (3marks)
- e) A table of values of a function *g* is given below:

<i>x</i>	0.0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6
<i>g</i> (<i>x</i>)	12.1	11.8	11.3	11.1	11.7	12.2	12.6	13.0	13.2

- Use Simpson's Rule to estimate: $\int_0^{1.6} g(x) dx$ (3marks)
- f) Use integration by parts to evaluate: $\int_1^2 \ln \sqrt{x+1} dx$ (4marks)
- g) The table below shows values of *x* and *y* for the function $y = \sqrt{x^3 + 1}$

<i>x</i>	0	0.5	1	1.5	2
$y = \sqrt{x^3 + 1}$	1.000		1.414		

- (i) Complete the table for the values of *y*. (1mark)
- (ii) Use the values in the table together with trapezoidal rule to estimate $\int_0^2 \sqrt{x^3 + 1} dx$ (3marks)

QUESTION TWO (TWENTY MARKS)

- a) Define a definite integral. (2marks)
- b) (i) Using trapezoidal rule with $n = 5$, estimate the value of the integral: $\int_2^5 x^2 - 3x + 2 dx$ (6marks)
- (ii) Find the exact value of: $\int_2^5 \frac{x^2 - 3x + 2}{x - 2} dx$ (3marks)
- (iii) Comment on your answers in b (i) and b (ii) above in relation to the number, *n*, of trapezoids used. (2marks)
- c) (i) Find an upper bound for the error in estimating: $\int_2^5 5x^2 dx$ using midpoint rule with $n = 5$. (4marks)

- (ii) How large should n (number of trapezoids) be in order to guarantee that midpoint approximation for: $\int_1^2 \frac{1}{x} dx$ is accurate to within 0.0001? (3marks)

QUESTION THREE (TWENTY MARKS)

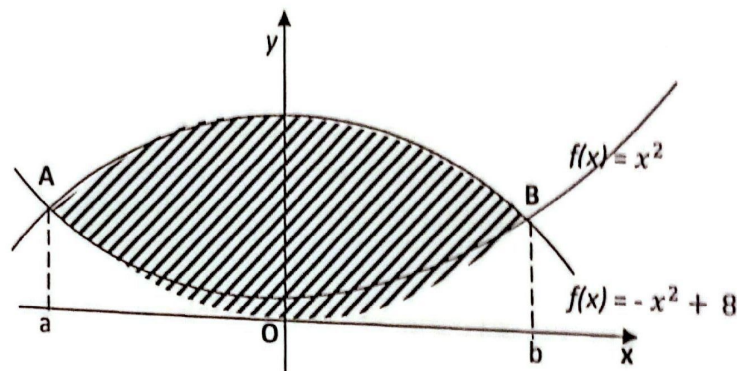
- a) Given that: $\frac{11x-1}{(1-x)^2(2+3x)} = \frac{A}{1-x} + \frac{B}{(1-x)^2} + \frac{C}{2+3x}$
- Find the values of A , B and C (7marks)
 - Find the exact value of: $\int_0^{\frac{1}{2}} \frac{11x-1}{(1-x)^2(2+3x)} dx$, giving your answer in the form $k + \ln a$, where k is an integer and a is a simplified fraction. (4marks)
- b) Suppose that t miles from the centre of a certain city the property tax revenue is approximately $R(t)$ thousand dollars per square mile, where $R(t) = 50e^{-\frac{t}{20}}$. Use this model to predict the total property tax revenue that will be generated by property within 10 miles of the city centre. (3marks)
- c) A growth company has its net earnings increase each year. Suppose that the earnings of the company at time t are being generated at the rate of $30 + 5t$ million dollars per year.
- Write a definite integral that gives the present value of the company's earnings over the next two years using a 10% interest rate. (2marks)
 - Compute the present value described in c (i) above. (4marks)

QUESTION FOUR (TWENTY MARKS)

- a) An integral is given as $\int_0^2 f(x) dx$, where $f(x) = \frac{1}{12}x^4 + 3x^2$
- Evaluate, to 5 decimal places, $\int_0^2 f(x) dx$ (2marks)
 - Make a rough sketch of the graph of $f''(x)$ for $0 \leq x \leq 2$ (4marks)
 - Find a number A such that $|f''(x)| \leq A$ for all x satisfying $0 \leq x \leq 2$ (2marks)
- b) (i) Approximate the value of the integral $\int_0^2 f(x) dx$ to 5 decimal places using midpoint rule with $n = 4$. (3marks)
- (ii) Compare your answers in a(i) and b(i) above. What is the percentage error due to use of midpoint approximations. (2marks)
- c) The line segment $x = 1 - y$, $0 \leq y \leq 1$, is revolved about the y -axis to generate a cone. Find the lateral surface area of the cone. (3marks)
- d) A company estimates that the rate of revenue produced by a machine at time t will be $5000 - 100t$ dollars per year. Find the present value of this continuous stream of income over the next four years at a 16% interest rate. (4marks)

QUESTION FIVE (TWENTY MARKS)

- a) A particle moves in a straight line. It passes through point O at $t = 0$ with a velocity $v = 5\text{m/s}$. The acceleration m/s^2 of the particle at time t seconds after passing through O is given by $a = 6t + 4$.
- Express the velocity, v of the particle at time t seconds in terms of t . (2marks)
 - Calculate the velocity of the particle when $t = 4$ (3marks)
 - Calculate the distance covered by the particle between $t = 1$ and $t = 4$ (5marks)
- b) The diagram below shows the sketch of the curve $f(x) = x^2$ and $f(x) = -x^2 + 8$ intersecting at points A and B .



- Find the value of a and b hence find the coordinates of A and B . (4marks)
- Find the area enclosed by $x = a$, $x = b$ the x -axis and the curves $f(x) = x^2$ and $f(x) = -x^2 + 8$ (6marks)





MULTIMEDIA UNIVERSITY OF KENYA

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

UNIVERSITY EXAMINATIONS 2020/2021

SECOND YEAR FIRST SEMESTER SUPPLEMENTARY/SPECIAL EXAMINATIONS

FOR THE DEGREE OF

BACHELOR OF SCIENCE IN COMPUTER SCIENCE/ BACHELOR OF SCIENCE IN

SOFTWARE ENGINEERING

CCS 2216: INTEGRAL CALCULUS

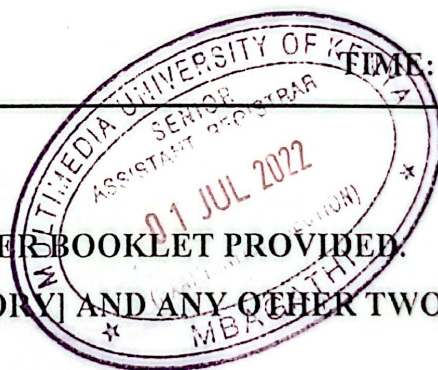
DATE: 15TH JUNE 2022

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS.



QUESTION ONE (THIRTY MARKS)

a) Determine;

i) $\int \left(\frac{3x^2 - 5x}{x} \right) dx$

(2 marks)

ii) $\int \frac{-3}{(2x-1)} dx$

(3 marks)

b) Evaluate each of the following integrals

i) $\int_0^{\frac{\pi}{6}} 24 \sin^5 \theta \cos \theta d\theta$

(4 marks)

ii) $\int_0^2 \frac{3x}{\sqrt{(2x^2+1)}} dx$

(4 marks)

Taking positive square roots only.

- c) Express $\frac{11-3x}{x^2+2x-3}$ in terms of partial fractions hence evaluate $\int \frac{11-3x}{x^2+2x-3} dx$ (6 marks)
- d) Determine the area enclosed between the curves $y = x^2 + 1$ and $y = 7 - x$ (6 marks)
- e) Evaluate $\int_0^1 5x e^{4x} dx$, correct to 3 significant figures (5 marks)

QUESTION TWO (TWENTY MARKS)

a) Show that;

$$\int \sqrt{(a^2 - x^2)} dx = \frac{a^2}{2} \sin^{-1} \frac{x}{a} + \frac{x}{2} \sqrt{(a^2 - x^2)} + c \text{ hence find the area of a}$$

circle whose equation is $x^2 + y^2 = 16$ (12 marks)

b) Show that;

$$\int \frac{1}{(a^2 + x^2)} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + c$$

hence evaluate $\int_0^3 \frac{5}{4+x^2} dx$

(8 marks)

QUESTION THREE (TWENTY MARKS)

a) Use the trapezoidal rule to evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{1+\sin x} dx$ using 6 intervals. Give the answer correct to 4 significant figures (8 marks)

b) i) Use Simpson's rule with 8 intervals to evaluate $\int_1^3 \frac{2}{\sqrt{x}} dx$, correct to 3 decimal places (9 marks)

ii) Evaluate the expression given in b(i) above analytically and compare the two answers obtained (3 marks)

QUESTION FOUR (TWENTY MARKS)

Using appropriate integration techniques, evaluate;

a) $\int_0^{\frac{\pi}{2}} x^2 \sin x dx$ (6 marks)

b) $\int_0^{\frac{\pi}{4}} 4 \cos^4 \theta d\theta$ (5 marks)

c) $\int \frac{5x^2 - 2x - 19}{(x+3)(x-1)^2} dx$

(5 marks)

d) $\int_0^1 \frac{3x}{(4x^2 - 1)^5} dx$

(4 marks)

QUESTION FIVE (TWENTY MARKS)

- a) Sketch the straight lines represented by $y = 4 - x$, $y = 3x$ and $3y = x$ hence determine by integration the area bounded by the three straight lines (6 marks)
- b) i) Determine the area enclosed by the two curves $y = x^2$ and $y^2 = 8x$ (6 marks)
- ii) if this area in (i) above is rotated 360° about the x-axis determine the volume of the solid of revolution produced (4 marks)
- c) The average value of a complex voltage waveform is given by;

$$V_{AV} = \frac{1}{\pi} \int_0^\pi (10 \sin \omega t + 3 \sin 3\omega t + 2 \sin 5\omega t) d(\omega t)$$

Evaluate A_{AV} correct to 2-decimal places

(4 marks)

